

Software Requirements Specifications

<Project Title>

Project Code:

Internal Advisor : Dr.**Muhammad** Ilyas

External Advisor: Dr. Muhammad Ilyas

Project Manager: Malaika Hamid

Project Team: Eisha Fatima
Minahil Fatima

Submission Date:

Signature **Document**

Project Manager's

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Definition of Terms, Acronyms and Abbreviations

This section should provide the definitions of all terms, acronyms, and abbreviations required to interpret the terms used in the document properly.

Term	Description
LFP	LOST AND Found portal
HTTP	Hyper text transfer protocol
TCP/IP	Transmission Control Protocol/Internet Protocol
SMTP	Simple mail transfer protocol

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Section 1

1. Introduction

1.1 Purpose of Document

I prepared the SRS document to describe all functional and non- functional requirements,for the lost and found portal. The SRS document is needed because it lists the features that the system must have so that a user can report, search for and claim lost items. The SRS document guides the development team .The intended readers of the SRS document are admin,software-developer,analyst and clients.

1.2 Project Overview

The Lost and Found Portal will be a web based system where people can easily report items they have lost, and likewise the results of what they find.The ultimate purpose of this mechanism is to assist users to get their lost items back, or to locate the owner if they have found something. It will also log all the reported items and let users search for them.

1.3 Scope

I wrote the scope to say what the system will do. I also wrote the scope to say what the system will not do.

What the System WILL Do: Let the users report an item, like a phone or keys. The users fill out a form that asks for the lost item name, the place the lost item was lost a picture of the lost item and other details. Let the users report a found item, with the details. Let the users search the database of reported lost and found items. The users can use keywords or filters such, as item type or location. Provide a process for a user to claim an item. The user believes the item belongs to the user. I need an Admin section. The system administrator can view, manage and verify all reported and claimed items. Send notifications (like an email) to users about their claim status.

What the system will not do: The system will not handle the exchange of the items. The system only gives a spot, for reporting and matching. The users must meet up or arrange the pickup or delivery themselves. The system will not have a payment gateway. Charge users any fees. The system is free to use. The system will not become an application. The system will only be a website that you open in a web browser.

2. Overall System Description

Describe the environment, in which the system will be developed and used, the anticipated users of the system and the known constraints, assumptions and dependencies.

2.1 User characteristics

The system is targeted to two main classes of users:

The Public: The most important set of users. It's someone who either lost something and they need to report it or found something and want to report it. They use the system to look for, report and claim articles.

Admin : Need (Controlling the system). Admin: Validates the reported and claimed items, Sends mail (or other message) to a user with updated status about his/her claims/cases(communities).

2.2 Operating environment

The Lost and Found Portal will be a regular web-based system.

Hardware Platform: The system will be based on a conventional client/server architecture. The users will be using the system via regular personal computing devices (laptops, smartphones, tablets).

Operating System

- I. Client-side: Any current operating system (like Windows, mac-OS, Android, iOS) which is compatible with a standard web browser.
- II. Server-side: An appropriate server operating system (like Linux or Windows Server).
- III. * Software Components/Applications: A regular web browser (like Chrome, Firefox, Safari) is the only client-side requirement. The server setup has to cooperate with the external email service for sending automated notifications.

2.3 System constraints

- I. **Software Constraints:** The system has to be a web-based website, and using any other platform such as mobile or desktop is not allowed for its implementation.
- II. **Hardware Constraints:** Any modern web-enabled devices are the only requirement to have no specific constraints above that.
- III. **Legal Constraints:** The physical custody or transfer of the lost or found items will not be processed by the system. Users will have to get together or arrange pick-up/delivery themselves.
- IV. **Environmental Constraints:** The system will be free for users, and there will be no incorporating any payment system or charging users for the service.
- V. **User constraints**
- VI. The project is aimed at the public that encompasses everyone, regardless of age and ability to use technology.
- VII. *Thus, the system will be equipped with a straightforward, user-friendly interface along with unambiguous instructions, placing user-friendliness above sophisticated technical controls.
- VIII. **Off-the-shelf components**
- IX. These are the components or existing software products that are used in the project, and they may bring their own limitations.
- X. The system relies only on a common web browser (such as Chrome, Firefox, and Safari) as its client-side requirement. This limitation requires the design to follow standard web development practices to ensure that it is compatible with all the major browsers.
- XI. The system will have to work with an outside email service for the purpose of sending notifications. The effectiveness and dependability of the notification feature will be a factor of the external email provider selected.

3. External Interface Requirements

This part is meant to explain the needs that guarantee the new Lost and Found Portal (LFP) system will connect to external components appropriately. It consists of hardware, software, and communication protocols essential for the system's efficient operation and interaction with the outside world.

Context Diagram (High-Level Abstraction)

The following high-level context diagram, as required, presents the LFP system and its main external interfaces.

- i. **User:** The origin of all inputs (lost/found reports, queries) and the destination of all outputs (search results, notifications).
- ii. **External Email Provider:** Necessary for transmitting reliable, transaction notifications.
- iii. **External Database:** Indispensable for persistent data storage and retrieval.

3.1 Hardware Interfaces

The LFP is an online application that relies on OS or standard network protocols to control all hardware interfaces indirectly.

- i. **1. User Devices (Client-Side):**
- ii. Description: A variety of computing devices such as desktop computers, laptops, tablets, and smartphones must be supported for interaction by the system.
- iii. Interaction: The standard input devices (keyboard, mouse, touch screen) and output devices (display) are mainly used for the interaction. The software gets the user input from these devices through the standard web browser interface.
- iv. Data/Control: The system should be capable of processing data from camera/photo inputs for uploaded item images. The LFP software uses standard HTML5 and responsive design to control its display across different screen resolutions.
- v. **2. Server Hardware:**
- vi. Description: The LFP software will be hosted on a typical server infrastructure could be physical or virtual.
- vii. Interaction: For all incoming and outgoing communication, the software will interact with the server's Network Interface Card (NIC) and will use the hard disk drive/SSD for data and log storage, which will be managed by the OS.
- viii. Data/Control: The database and image storage will need high-speed and reliable access.

ix. Software Interfaces

The LFP is connected to the following external software components:

Component , Example Version , Purpose & Service Provided , Data Exchanged

Relational Database : Controls the persistent storage of all LFP data (users, items, reports). Guarantees data persistence services. SQL queries (perform operations), result sets with user profile data, and item details.

External Email Service : Allows system-generated notifications and alerts to users to be sent through a highly reliable mechanism. | Recipient email address, subject line, email body content (HTML/plain text).

Operating System : The basic runtime environment is made available as well as access to the file system for image storage and management of resources. File paths, I/O requests for image uploads and retrieval, system logs.

Web Browsers , Chrome, Firefox, Safari (Latest) . The user has to install this mandatory client-side software in order to access the portal interface. HTTP/HTTPS requests (user input, search criteria, credentials), HTTP/HTTPS responses (HTML, JSON, CSS, images).

3.2 Communications Interfaces

The LFP system employs standard internet protocols completely for the purpose of communication between the server, external services, and the users.

1. TCP/IP (Transmission Control Protocol/Internet Protocol):

Function: The basic protocols for network communication, allowing reliable and ordered delivery of data across the network.

2. **Application:** Employed for all types of connectivity, which includes client-server interaction and server-to-external service communication like database, email.

3. HTTPS (Hypertext Transfer Protocol Secure)

Function: The essential application-layer protocol for all data transferring between the User's Web Browser and the LFP Server. It guarantees that all communications are encrypted through the use of TLS/SSL.

Requirement: Login Name and Password, lost/found reports, and PII have to be sent via HTTPS only (Port 443).

4. SMTP (Simple Mail Transfer Protocol):

Function: The protocol that the External Email Service uses to send notifications from the system to the end-user mailboxes.

4. Functional Requirements

Functional requirements are a description of what the system or any of its parts must necessarily do through functions. They provide the answer to what the system must implement in order to meet the business requirements along with user requirements.

1.General structure:

A well-described functional requirement tends to follow this pattern:

Element ,Description .

ID: Unique identifier for subsequent reference (e.g., FR-001)

Description : A precise description of the needed action of the system. The system needs to give a registered user the option to change their password through email.

Business Event : The event or step in the process this requirement is concerned with. User cannot access account (Forgot Password)

2. Place Emphasis on Business Events and Processes

Your writing has pointed out that Functional Requirements must be contextualized in terms of business events and the business processes.

Business Event: A situation that necessitates a reaction from the system.

Examples: A customer places an order, a user tries logging in, an inventory item's stock level goes below a certain point, a payment gets approved.

Business Process: A set of actions or steps that contribute to the achievement of a business goal when executed.

Example: The Order Fulfillment Process may include the events: Place Order Check Inventory Process Payment Ship Item.

5. Non-functional Requirements

5.1 Performance Requirements

The system shall be able to perform in accordance with the following characteristics:

Speed/Response Time: For a standard query during maximum use, the system will show the results of searching for lost or found items within 2 seconds. The homepage and static pages will be loaded in 1 second.

Capacity: The portal will allow at least 50 users to access it at the same time without any performance loss. It should also be able to accommodate up to 500 new item reports a day.

Precision: Information on all reported items (date, time, place) will be preserved and displayed with the utmost precision. Location data will either be expressed in standard geographical coordinates (if applicable) or in detailed verbal description.

Reliability: The system will be guaranteed an uptime of not less than 99.9% during maximum hours . The maximum time for scheduled maintenance should be 2 hours per month.

Availability: The portal will be accessible round the clock all week long excluding the time of scheduled maintenance.

5.2 Safety Requirements

The configuration of the system should be such that it safeguards against loss, damage, and harm, especially with regard to data of users and their safety (in case of item return meetings).

Data Integrity: The system will have to establish a procedure of daily backups for all the item data and user data reported in order to avert any loss caused by a system crash.

Prevention of Misuse: The portal will require user verification (for instance, by confirming through email or phone number) in order to reduce the occurrences of fake reports and the chances of malicious activities.

Guidance for Item Exchange: When a user is arranging to return a found item, the system should show a safety disclaimer and indicate to the user that they should meet at public, safe places (like police stations or certified campus security zones).

Certification/Policies: The system will have to follow with the applicable data protection regulations concerning the retention and showing of personal data.

5.3 Security Requirements

The system will fulfill these requirements in terms of security, integrity, and data protection:

Authentication & Authorization:

All users should go through a safe login process (username/password or social login) for authentication.

Only the user who reported the found/lost item is allowed to edit or delete the report.

An administrative module must be implemented with role-based access control (RBAC) for managing both users and content.

Data Protection & Privacy:

Personal data such as passwords, names, emails, and phone numbers must be encrypted during transmission and when stored (in the database).

The portal will follow the data minimization principle and will only provide necessary contact details (e.g. a secure, hidden contact method) for the item's return.

Integrity: The system will record all sensitive transactions like item deletions, status modifications, and admin actions to guarantee an audit trail and ensure data integrity.

1.1 8.5 Business Rules

1.1 List any operating principles about the product, such as which individuals or roles can perform which functions under specific circumstances. These are not functional requirements in themselves, but they might imply certain functional requirements to enforce the rules. Mention all users who will be accessing the software and describe their respective rights.

1.1 -

5.4 User Documentation

User Manual: An all-inclusive document that covers all functionalities, informs on how to work with the system, and gives system maintenance tasks. It will have detailed instructions for all important processes step-by-step.
Online Context-Sensitive Help: Help that is integrated and that can be accessed through the application interface, which provides immediate and relevant assistance for the specific screen or function that is currently being viewed.
Installation/Deployment Guide: A document written for IT staff or administrators that lays out the steps for installing, configuring, and deploying the software in the selected environment.
Quick Start Guide/Tutorial: A short introductory guide that will allow new users to be quickly up and running with the basic features of the system.

6. Assumptions and Dependencies

Assumptions:

Availability of Third-Party : It is assumed that all necessary external APIS or services (e.g., payment gateways, external data feeds) required for system integration will be available, stable, and documented in a timely manner.

User Technical Proficiency: It is assumed that end-users possess basic computer literacy and are familiar with standard web browser functionality.

Hardware and OS Compatibility: The target environment will meet the minimum hardware and operating system specifications outlined in the Installation Guide.

Dependencies:

Database Infrastructure: The project is dependent on the IT department's successful setup and provisioning of the dedicated production and staging database environments by the defined project milestone.

External Component Development: The system's integration module is dependent on the timely and successful delivery of the 'Inventory Management Service' component being developed by the internal 'Operations' team.

Security Audit Completion: Final deployment is dependent on the successful completion and sign-off of a mandatory security audit by the Security & Compliance team.

7. References

This section should provide a complete list of all documents referenced at specific point in time. Each document should be identified by title, report number (if applicable), date, and publishing organization. Specify the sources from which the references can be obtained. (This section is like the bibliography in a published book).

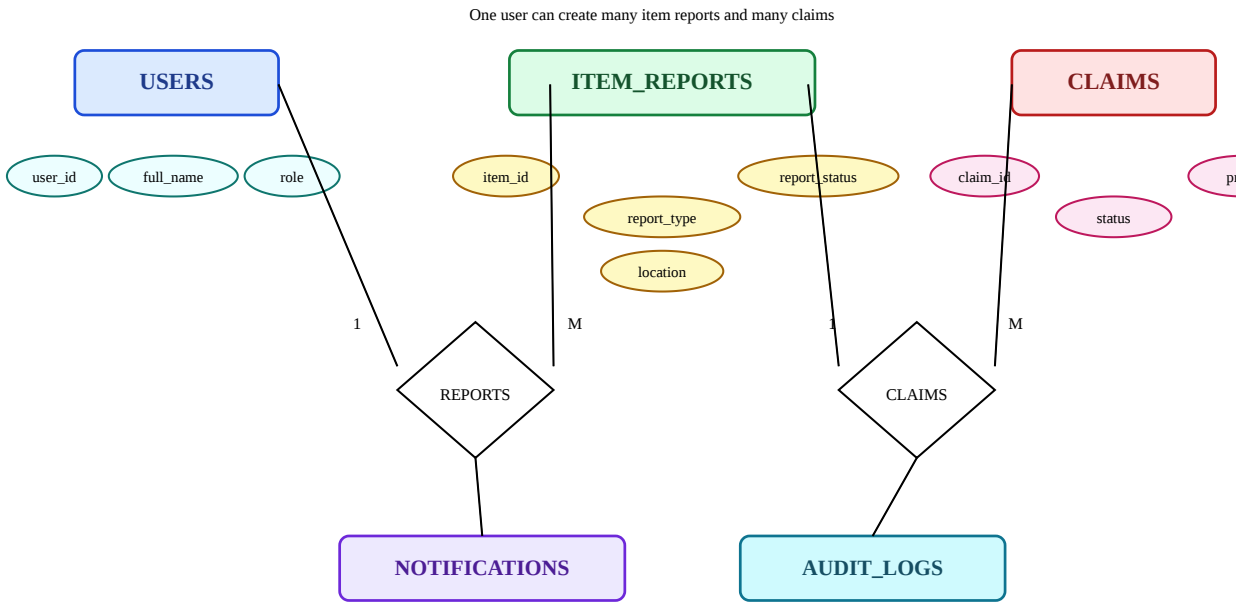
Ref. No.	Document Title	Date of Release/ Publication	Document Source	
PGBH01-2003-Proposal	Project Proposal	Oct 20, 2003	<Give the path of your Project repository/Folder>	
PGB-01-2023	PROJECT PROPOSAL	OCT 20,2023	Project management office repository	
SDD-v1.0	Software design document	Nov 15,2023d	Development team share-point/drive	

8. Appendices

This section should include supporting detail that would be too distracting to include in the main body of the document.

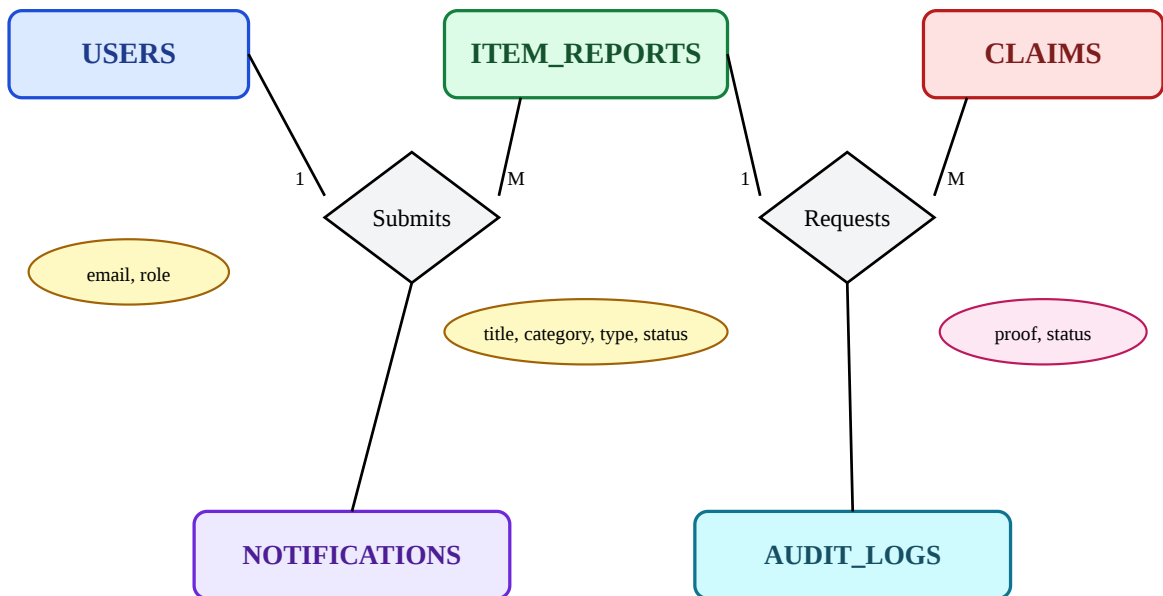
8.1 ER Diagram - Option 1 (Complete Project View)

This diagram covers full project entities: Users, Item Reports, Claims, Notifications, and Audit Logs.



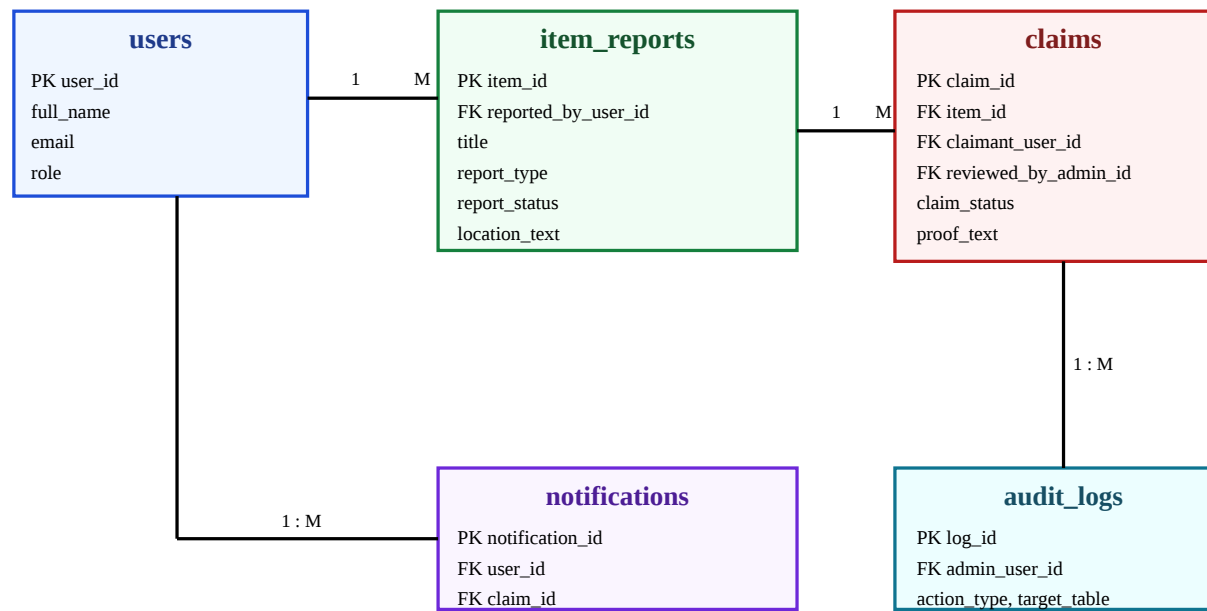
Option 1 Explanation: This full project ER view shows five main entities: USERS, ITEM_REPORTS, CLAIMS, NOTIFICATIONS, and AUDIT_LOGS. The REPORTS relationship connects users with item reports (one user can create many reports). The CLAIMS relationship connects item reports with claims (one report can receive many claim requests). Notifications are linked for status updates, and audit logs track admin-level actions for control and transparency.

8.2 ER Diagram - Option 2 (Transaction Focus)



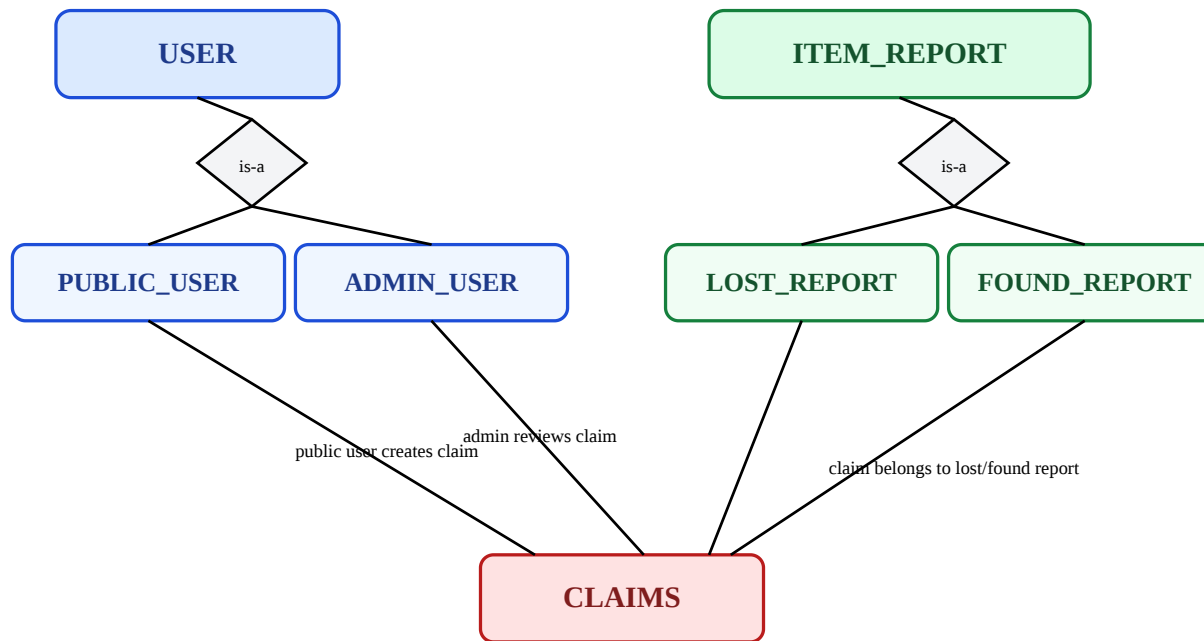
Option 2 Explanation: This diagram focuses on transaction flow. A user submits an item report and users request claims on available reports. The central part explains movement from reporting to claim request. NOTIFICATIONS deliver claim status messages to users, while AUDIT_LOGS keep a history of reviewed actions. This diagram is useful to explain process flow in presentation.

8.3 ER Diagram - Option 3 (Relational Mapping)



Option 3 Explanation: This diagram maps ER directly to database tables. It clearly shows primary keys and foreign keys. The users table links to item_reports through reported_by_user_id. The claims table links to both users and item_reports. notifications links to users and claims. audit_logs links to admin users. This is the best view to explain how ER design is implemented in SQL tables.

8.4 Extended ER Diagram (EER)



EER Explanation: The EER diagram adds specialization. USER is divided into PUBLIC_USER and ADMIN_USER (role-based behavior). ITEM_REPORT is divided into LOST_REPORT and FOUND_REPORT (type-based behavior). CLAIMS stays central because claim requests are created by public users and reviewed by admins. This makes project logic clearer than a basic ER model.

8.5 Diagram Explanation (Simple)

- 1) USERS stores all people of the portal. Their role decides whether they are public users or admins.
- 2) ITEM_REPORTS stores each lost/found report with type, status, and location details.
- 3) CLAIMS stores ownership requests made by users for a specific reported item.
- 4) NOTIFICATIONS sends status updates to users. AUDIT_LOGS keeps admin activity history.
- 5) In EER, USER is specialized into PUBLIC_USER and ADMIN_USER, and ITEM_REPORT is specialized into LOST_REPORT and FOUND_REPORT.

8.6 SQL Table Creation (XAMPP MySQL)

```

CREATE DATABASE IF NOT EXISTS lost_found_portal;
USE lost_found_portal;

CREATE TABLE users (
  user_id INT AUTO_INCREMENT PRIMARY KEY,
  full_name VARCHAR(100) NOT NULL,
  email VARCHAR(120) NOT NULL UNIQUE,
  phone VARCHAR(20),
  password_hash VARCHAR(255) NOT NULL,
  role ENUM('PUBLIC', 'ADMIN') NOT NULL DEFAULT 'PUBLIC',
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE item_reports (
  item_id INT AUTO_INCREMENT PRIMARY KEY,
  title VARCHAR(120) NOT NULL,
  description TEXT,
  category VARCHAR(60) NOT NULL,
  color VARCHAR(40),
  report_type ENUM('LOST', 'FOUND') NOT NULL,
  report_status ENUM('OPEN', 'MATCHED', 'CLAIMED', 'CLOSED') NOT NULL DEFAULT 'OPEN',
  location_text VARCHAR(150) NOT NULL,
  event_date DATE NOT NULL,
  photo_url VARCHAR(255),
  reported_by_user_id INT NOT NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (reported_by_user_id) REFERENCES users(user_id)
);

CREATE TABLE claims (
  claim_id INT AUTO_INCREMENT PRIMARY KEY,
  item_id INT NOT NULL,

```

```
claimant_user_id INT NOT NULL,
proof_text TEXT,
claim_status ENUM('PENDING','APPROVED','REJECTED') NOT NULL DEFAULT 'PENDING',
reviewed_by_admin_id INT NULL,
claimed_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
reviewed_at TIMESTAMP NULL,
FOREIGN KEY (item_id) REFERENCES item_reports(item_id),
FOREIGN KEY (claimant_user_id) REFERENCES users(user_id),
FOREIGN KEY (reviewed_by_admin_id) REFERENCES users(user_id)
);

CREATE TABLE notifications (
notification_id INT AUTO_INCREMENT PRIMARY KEY,
user_id INT NOT NULL,
claim_id INT NULL,
message_text VARCHAR(255) NOT NULL,
channel ENUM('EMAIL','SMS','IN_APP') NOT NULL DEFAULT 'EMAIL',
is_sent TINYINT(1) NOT NULL DEFAULT 0,
sent_at TIMESTAMP NULL,
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
FOREIGN KEY (user_id) REFERENCES users(user_id),
FOREIGN KEY (claim_id) REFERENCES claims(claim_id)
);

CREATE TABLE audit_logs (
log_id INT AUTO_INCREMENT PRIMARY KEY,
admin_user_id INT NOT NULL,
action_type VARCHAR(80) NOT NULL,
target_table VARCHAR(60) NOT NULL,
target_id INT NOT NULL,
action_note VARCHAR(255),
action_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
FOREIGN KEY (admin_user_id) REFERENCES users(user_id)
);
```

8.7 Insert Commands (Sample Rows)

```
INSERT INTO users (full_name, email, phone, password_hash, role) VALUES
('Ali Raza','ali@example.com','03001234567','hash_ali_123','PUBLIC'),
('Sara Khan','sara@example.com','03111234567','hash_sara_123','PUBLIC'),
('Hamza Iqbal','hamza@example.com','03221234567','hash_hamza_123','PUBLIC'),
('Admin One','admin@example.com','03331234567','hash_admin_123','ADMIN');

INSERT INTO item_reports (title, description, category, color, report_type, report_status, location_text, event_date,
photo_url, reported_by_user_id) VALUES
('Black Wallet','Leather wallet with student card','Wallet','Black','LOST','OPEN','Main Gate','2026-03-20','wallet.jpg',1),
('Silver Watch','Round dial wrist watch','Watch','Silver','FOUND','OPEN','Library Hall','2026-03-22','watch.jpg',2),
('Blue Backpack','Bag with books and notebook','Bag','Blue','LOST','MATCHED','Cafeteria','2026-03-18','bag.jpg',3),
('Car Keys','Two keys with red keychain','Keys','Gray','FOUND','OPEN','Parking Area','2026-03-25','keys.jpg',2);

INSERT INTO claims (item_id, claimant_user_id, proof_text, claim_status, reviewed_by_admin_id, reviewed_at) VALUES
(2,1,'I can identify the scratch on the watch strap.','PENDING',NULL,NULL),
(3,1,'The backpack has my name tag inside.','APPROVED',4,NOW()),
(4,3,'The keychain is red and has letter H.','REJECTED',4,NOW()),
(1,2,'Wallet contains my card number ending 52.','PENDING',NULL,NULL);
```

8.8 Data Display After INSERT

```
SELECT user_id, full_name, role FROM users;
SELECT item_id, title, report_type, report_status FROM item_reports;
SELECT claim_id, item_id, claimant_user_id, claim_status FROM claims;
```

item_id	title	report_type	report_status
1	Black Wallet	LOST	OPEN
2	Silver Watch	FOUND	OPEN
3	Blue Backpack	LOST	MATCHED
4	Car Keys	FOUND	OPEN

Result: SQL commands, insert commands, and query output are now formatted with the same document-style background and clear spacing.